Q03 Deeper Project Suggestions

1. **Reward hacking demo:** train a tiny reward model on synthetic preferences with a known blind spot, then run PPO-style optimization against it; document the exploit the policy finds and how the KL anchor changes its severity.
2. **DPO identity check:** implement the derivation numerically — verify on a bandit toy that the DPO gradient equals the RLHF-objective gradient through the closed-form optimum, and characterize when finite samples break the equivalence.
3. **GRPO stress test:** implement group-relative advantages on a curriculum of tasks with varying solve rates; plot advantage variance and update quality as the group becomes all-success or all-failure.
4. **Length-bias audit:** take a public preference dataset; measure the correlation between response length and preference label; then test whether a DPO-trained small model inherits the bias.